STRUCTURAL STUDY OF THE TOLC CHANNEL OF THE ACRAB-TOLC EFFLUX PUMP IN EXTRACELLULAR VESICLES

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ABSTRACT

The AcrAB-TolC efflux pump of *Escherichia coli* contributes significantly to antibiotic resistance, but its structural study is limited by detergent-based solubilization workflows that can destabilize proteins and remove their native lipid environment. Here, we present a proof-of-concept approach using extracellular vesicles (EVs) for cryo-EM analysis of membrane proteins in a near-physiological context.

Controlled overexpression of TolC in *E. coli* C43(DE3) induces the production of outer membrane vesicles. Single-particle cryo-EM of purified vesicles yielded a 4.16 Å structure of TolC in its closed conformation, together with an open conformation at 5.62 Å, a state previously observed only in the presence of partner proteins. This suggests that the periplasmic domain of TolC opens spontaneously, with partner binding subsequently stabilizing the open state, rather than triggering it.

Ongoing work aims to improve resolution with a larger dataset, extend the method to OprM and OpmQ from *Pseudomonas aeruginosa*, and ultimately study the complete AcrAB-TolC complex in vesicles.

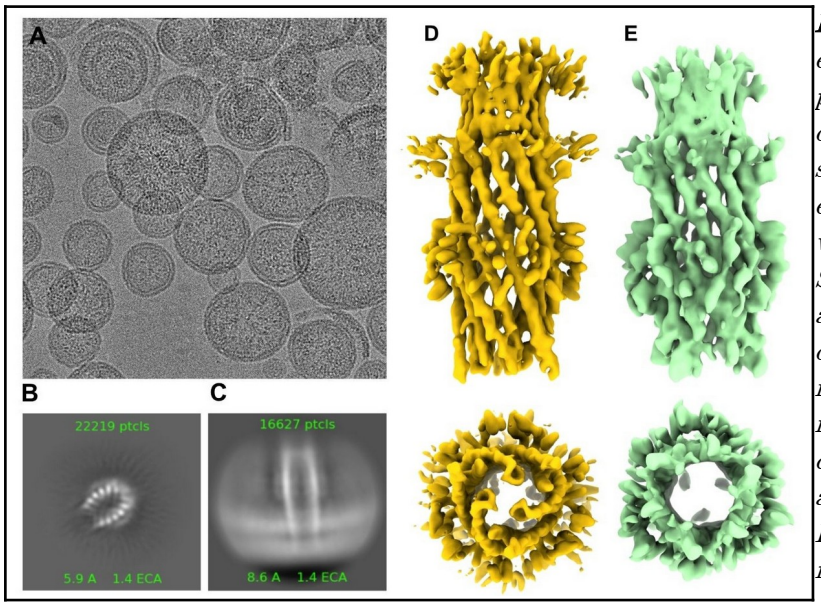


Fig. *Figure 2: Cryo-EM analysis of TolC in extracellular vesicles. A: Cryo-EM image of purified vesicles obtained following the overexpression of TolC in the *E. coli* C43 strain. B and C: 2D classes of TolC in extracellular vesicles, showing top and side views of TolC, respectively. D and E: Structure of TolC in extracellular vesicles in a closed conformation (yellow, D) and an open conformation (green, E), with resolutions of 4.16 Å and 5.62 Å, respectively. Results obtained from a dataset of 22,000 micrographs collected on an ESRF Titan microscope equipped with a Falcon 4i detector and a Selectris X energy filter.*

REFERENCES